

Coursework Assignment Questions (Set 4/4)

Due date: November 11, 2024

Please submit your answers to all questions as a single file for Coursework 1. The C++ program code and the report (in MS Word or PDF format) should be uploaded to Blackboard. If the program source code is not submitted either as a separate file or included within the report, the exercise will be marked as 'not attempted' (i.e., zero marks), even if numerical results and discussions are provided. All source code must be copied and pasted into the document—screenshots will not be accepted.

Set 3: One question of 25 marks

Q6 (Marks 25): C++ Variable Manipulation and Output.

Write C++ statements that accomplish the following tasks. Ensure that you understand each task and can explain your code.

Tasks:

1) Declare and Initialize Variables

- Declare two integer variables, `x` and `y`. Initialize `x` to 25 and `y` to 18.
- Declare and initialize a double variable named `payRate` to 12.50.
- Create an integer variable called `firstNum` and initialize it to a value of your choice. Then, declare another integer variable `tempNum` and copy the value of `firstNum` into `tempNum`.

2) Swap Values

Write the code to swap the contents of the integer variables `x` and `y`. You may declare additional variables as necessary. Explain how your swap works in a comment.

3) Output Expressions

Assume `x` and `y` are now double variables. Write a C++ statement to output the contents of `x`, `y`, and the result of the expression $x + 12 / y - 18 + x^3$.

Explain the output in a comment above the line of code.

Submission Instructions:

- Write your code in a `.cpp` file.
- Provide explanations for each task as comments in your code.
- Ensure your code compiles without errors.
- Avoid using AI tools for code generation. Your submission will be checked for originality, and any signs of AI usage will be flagged.

Total: 25 marks for the programming exercise

Functionality (15 marks)

1. Non-Functional Code:
Some evidence of programming, but the code does not compile or run: max 3 marks.
2. Functional Code with Errors:
The program compiles and runs, but the results are incorrect or inconsistent: max 3 marks.
3. Partially Correct Functionality:
The program works, and the results are partially correct or complete: max 3 marks.
4. Fully Correct Functionality:
The program works correctly, and the results are accurate and complete: max 3 marks.
5. Exceptional Functionality:
The program is exceptionally well done, featuring extensions and/or additional functionalities:
max 3 marks.

Code Quality (10 marks)

Well-Described and Commented Code:

The code is well-structured, with clear, concise comments explaining the logic and flow of the program.